

ODD 2024
Tutorial Sheet – 4 & 5 (Solutions)
Software Development Fundamentals – I (15B11CI111)

COURSE OUTCOMES		COGNITIVE LEVELS
CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts	Understand (Level 2)
CO2	Explain basics of C programming concepts to make decision for solving problems	Understand (Level 2)
CO3	Demonstrate and contrast different methods for writing modular programs in C	Understand (Level 2)
CO4	Use various C programming constructs to implement iteration, and recursion	Apply (Level 3)
CO5	Apply and implement arrays, pointers, structures and file handling for solving real-world problems	Apply (Level 3)

Note: Students are advised to submit their solutions to their respective tutorial faculties

Q1. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    int i;
    for(i = 0; i>9; i+=3)
    {
        printf("for ");
    }
    return 0;
}
```

Output: Nil (due to the condition)

Q2. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    int i = 1, j = 1;
    for(--i && j++ ; i<10; i+=2)
    {
        printf("loop ");
    }
    return 0;
}
```

Output: loop loop loop loop loop

Q3. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    for(5;2;2)
        printf("Hello");
    return 0;
}
```

Output: Infinite times Hello is printed (due to condition)

Q4. What will be the output of the C program?

```
#include<stdio.h>
```

```
int main()
{
    int n;
    for (n = 9; n!=0; n--)
        printf("n = %d", n--);
    return 0;
}
```

Output: Infinite (due to condition and decrement criteria)

Q5. What will be the output of the C program?

```
#include <stdio.h>
int main()
{
    int c = 5, no = 10;
    do {
        no /= c;
    } while(c--);

    printf ("%dn", no);
    return 0;
}
```

Output: Nil (due to condition and decrement criteria)

Q6: Write C programs to print the following patterns:

(a) <pre> *</pre>	(b) <pre> 1 2 3 4 5 1 2 3 4 1 2 3 1 2 1</pre>
(c) <pre> A B C D E F G H I A B C D E F G A B C D E A B C A</pre>	(d) <pre> *</pre>

(a)

// C program to print right half pyramid pattern of star

```
#include <stdio.h>
```

```
int main()
{
    int rows = 5;

    // first loop for printing rows
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j <= i; j++) {
            printf("* ");
        }
    }
}
```

```

*
* *
* * *
* * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```

    }
    printf("\n");
}
return 0;
}

```

(b)

// C program to print the inverted left half pyramid pattern of numbers
#include <stdio.h>

```

int main()
{
    int rows = 5;

    // first loop for printing all rows
    for (int i = 0; i < rows; i++) {

        // first inner loop for printing white spaces
        for (int j = 0; j < 2 * i; j++) {
            printf(" ");
        }

        // second inner loop for printing numbers
        for (int k = 0; k < rows - i; k++) {
            printf("%d ", k + 1);
        }
        printf("\n");
    }

    return 0;
}

```

```

1 2 3 4 5
  1 2 3 4
    1 2 3
      1 2
        1

```

(c)

// C program to print the inverted full pyramid pattern of alphabets
#include <stdio.h>

```

int main()
{
    int rows = 5;

    // first loop for printing all rows
    for (int i = 0; i < rows; i++) {

        // first inner loop for printing leading white
        // spaces
        for (int j = 0; j < 2 * i; j++) {
            printf(" ");
        }

        // second inner loop for printing alphabets
        for (int k = 0; k < 2 * (rows - i) - 1; k++) {
            printf("%c ", 'A' + k);
        }
        printf("\n");
    }
}

```

```

A B C D E F G H I
  A B C D E F G
    A B C D E
      A B C
        A

```

(d)

// C Program to print hourglass pattern using star *

```

#include <stdio.h>
int main()
{
    int rows = 5;

```

```

    // first outer loop to iterate each row

```

```

* * * * *
 * * * * *
  * * * *
   * * *
    *
   * * *
  * * * *
 * * * * *
* * * * *

```

```

for (int i = 0; i < 2 * rows - 1; i++) {

    // assigning comparator
    int comp;
    if (i < rows) {
        comp = 2 * i + 1;
    }
    else {
        comp = 2 * (2 * rows - i) - 3;
    }

    // first inner loop to print leading spaces
    for (int j = 0; j < comp; j++) {
        printf(" ");
    }

    // second inner loop to print star *
    for (int k = 0; k < 2 * rows - comp; k++) {
        printf("* ");
    }
    printf("\n");
}
return 0;
}

```

Q7. Write C Programs to do the following:

- (i) Write a program in C language for printing all the odd / even numbers within a given range using the while loop.

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    int num1, num2,r,i;

    printf("Enter the first number for the range: ");
    scanf("%d",&num1); //received the number for num1
    printf("Enter the second number for the range: ");
    scanf("%d",&num2); //received the number for num2
    printf("\nDisplay the even numbers between %d and %d are: ",num1,num2);

    for(i=num1; i<=num2; i++){
        r=i%2;
        if(r==0)
            printf("\n %d",i);
    }

    printf("\n\nDisplay the odd numbers between %d and %d are: ",num1,num2);

    for(i=num1; i<=num2; i++){
        r=i%2;
        if(r==1)
            printf("\n %d",i);
    }
    getch();
    return 0;
}

```

- (ii) Write a program in C language to print the addition of numbers within a range.

```

#include <stdio.h>
#include <conio.h>

```

```

void main()
{
    int num, i, sum = 0; // declare local variables
    printf(" Enter a positive number: ");
    scanf("%d", &num); // take any positive number
    // executes until the condition remains true.
    for (i = 0; i <= num; i++)
    {
        sum = sum + i; // at each iteration the value of i is added to the sum variable
    }
    // display the sum of natural number
    printf("\n Sum of the first %d number is: %d", num, sum);
    getch();
}

```

- (iii) Write a program in C language to print the factorial of a number.

```

#include<stdio.h>
int main()
{
    int i,fact=1,number;
    printf("Enter a number: ");
    scanf("%d",&number);
    for(i=1;i<=number;i++){
        fact=fact*i;
    }
    printf("Factorial of %d is: %d",number,fact);
    return 0;
}

```

- (iv) Write a program in C language to print all the uppercase letters.

```

#include <stdio.h>

int main()
{
    char i;

    printf("Capital (upper) case characters:\n");
    for(i='A'; i<='Z'; i++)
        printf("%c ",i);

    printf("\n\nLower case characters:\n");
    for(i='a'; i<='z'; i++)
        printf("%c ",i);

    return 0;
}

```

- (v) Write a program in C language to print the multiplication table of an integer.

```

#include <stdio.h>
int main() {
    int n;
    printf("Enter an integer: ");
    scanf("%d", &n);

    for (int i = 1; i <= 10; ++i) {
        printf("%d * %d = %d \n", n, i, n * i);
    }
    return 0;
}

```

- (vi) Write a program to print the square roots of all the numbers within a range.

```
#include <stdio.h>
#include <math.h>

int main()
{
    int i,n;

    printf("Enter the value of N: ");
    scanf("%d",&n);

    printf("No      Square Root\n",n);
    for(i=1;i<=n;i++)
    {
        printf("%d \t %.2f\n",i,sqrt((double)i));
    }

    return 0;
}
```

- (vii) Write a program in C to print a range of numbers (between **a** to **b**) that are divisible by 5.

```
#include <stdio.h>

void main()
{
    int i, num1, num2, count = 0, sum = 0;

    printf("Enter the value of num1 and num2 \n");
    scanf("%d %d", &num1, &num2);
    /* Count the number and compute their sum*/
    printf("Integers divisible by 5 are \n");
    for (i = num1; i < num2; i++)
    {
        if (i % 5 == 0)
        {
            printf("%3d,", i);
            count++;
            sum = sum + i;
        }
    }
    printf("\n Number of integers divisible by 5 between %d and %d = %d\n", num1, num2, count);
    printf("Sum of all integers that are divisible by 5 = %d\n", sum);
}
```

- (viii) Write a program in C to print arithmetic progression and its sum.

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int first, diff, terms, value, sum=0, i;

    printf("Enter the number of terms in AP series\n");
    scanf("%d", &terms);

    printf("Enter first term and common difference of AP series\n");
    scanf("%d %d", &first, &diff);

    /* print the series and add all elements to sum */
    value = first;
```

```

printf("AP SERIES\n");
for(i = 0; i < terms; i++) {
    printf("%d ", value);
    sum += value;
    value = value + diff;
}

printf("\nSum of the AP series till %d terms is %d\n", terms, sum);

return 0;
}

```

(ix) Write a C program to print all Armstrong Numbers from 1 to N.

```

#include <stdio.h>

/*function to check Armstrong Number */
int checkArmstrong(int num)
{
    int tempNumber, rem, sum;
    tempNumber = num;

    sum = 0;
    while (tempNumber != 0) {
        rem = tempNumber % 10;
        sum = sum + (rem * rem * rem);
        tempNumber /= 10;
    }
    /* checking number is Armstrong or not */
    if (sum == num)
        return 1;
    else
        return 0;
}

int main()
{
    int i, n;

    printf("Enter the value of N: ");
    scanf("%d", &n);

    printf("All Armstrong numbers from 1 to %d:\n", n);
    for (i = 1; i <= n; i++) {
        if (checkArmstrong(i))
            printf("%d,", i);
    }

    return 0;
}

```